Factuality Annotation Guidelines

LLM Factuality Evaluation Study

Background & Purpose

Thank you for contributing to this study. We genuinely appreciate your time!

We are studying how well large language models (LLMs) communicate scientific information. Specifically, we want to know whether LLM outputs are factually accurate. To do this, we use automated metrics, but they need to be validated against human judgment. That is where you come in. We will use your annotations to see if there is a correlation between automatic metrics and human judgment to understand if these metrics can be used in this context.

You will evaluate individual claims extracted from LLM outputs in your assigned scientific domain. We are evaluating five main domains:

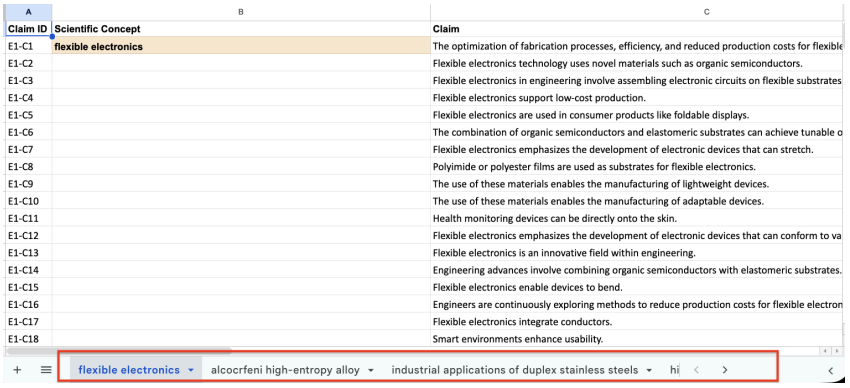
1. Engineering
2. Life Sciences
3. Physical Sciences and Mathematics
4. Social and Behavioural Sciences
5. Arts and Humanities

What You Will See

From each domain, we are annotating a subset of 10 scientific concepts we asked LLMs about.

You will receive two links:

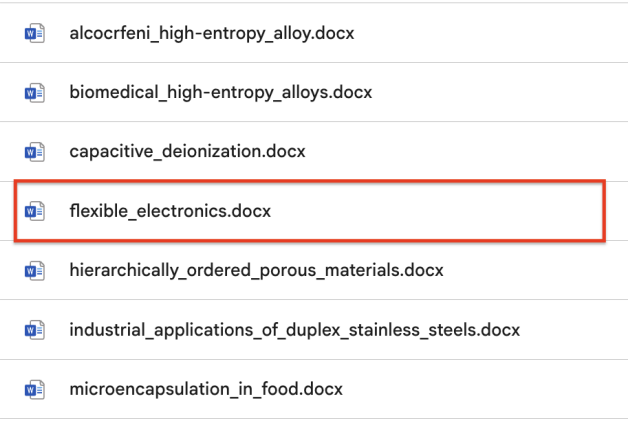
- A spreadsheet with 10 sheets, each one containing claims about a different concept in your domain. You can navigate through the concepts at the bottom of the spreadsheet:



Claim ID	Scientific Concept	Claim
E1-C1	flexible electronics	The optimization of fabrication processes, efficiency, and reduced production costs for flexible
E1-C2		Flexible electronics technology uses novel materials such as organic semiconductors.
E1-C3		Flexible electronics in engineering involve assembling electronic circuits on flexible substrates
E1-C4		Flexible electronics support low-cost production.
E1-C5		Flexible electronics are used in consumer products like foldable displays.
E1-C6		The combination of organic semiconductors and elastomeric substrates can achieve tunable o
E1-C7		Flexible electronics emphasizes the development of electronic devices that can stretch.
E1-C8		Polyimide or polyester films are used as substrates for flexible electronics.
E1-C9		The use of these materials enables the manufacturing of lightweight devices.
E1-C10		The use of these materials enables the manufacturing of adaptable devices.
E1-C11		Health monitoring devices can be directly onto the skin.
E1-C12		Flexible electronics emphasizes the development of electronic devices that can conform to va
E1-C13		Flexible electronics is an innovative field within engineering.
E1-C14		Engineering advances involve combining organic semiconductors with elastomeric substrates.
E1-C15		Flexible electronics enable devices to bend.
E1-C16		Engineers are continuously exploring methods to reduce production costs for flexible electron
E1-C17		Flexible electronics integrate conductors.
E1-C18		Smart environments enhance usability.

flexible electronics | alcocrfeni high-entropy alloy | industrial applications of duplex stainless steels | hi

- A Google Drive folder containing one reference document (.docx) per scientific concept in your domain. For example, this is the corresponding evidence document to the concept shown above:



Each row in each sheet contains:

- The scientific concept that the LLM output is about
- A claim ID (e.g. E1-C3), which can be used to find the corresponding sources in the reference document
- An atomic claim about the concept. (An atomic claim is a single, self-contained statement extracted from an LLM output.)

Annotation Workflow

For each concept:

1. Open the matching reference document from the Google Drive folder. At the top of each document, you will find a link to the survey paper associated with that concept, followed by the search snippets for each claim, organised by claim ID. This is the **evidence** which you will use to decide the factuality of the claim.
2. Go through each claim and its evidence and label (1) scientific check-worthiness and (2) factuality, as will be explained below.

Each concept has a maximum of 20 claims. We expect each claim to take about 1 minute and thus the total time to be approximately 3 hours, which you can complete at your own pace.

Label 1: Is This Claim Scientifically Check-Worthy?

A check-worthy scientific claim is a statement that can be verified against the scientific literature. This includes:

> **✓ Check-worthy — Historical fact**
> *James Watson and Francis Crick discovered the double-helix structure of DNA in 1953*

> **✓ Check-worthy — Method**
> *Randomized controlled trials are used to establish causal relationships between interventions and outcomes.*

✓ **Check-worthy — Prediction / causal claim**
The use of electronic devices for reading can lead to increased distractibility.

Not all claims can be fact-checked. In particular, vague statements that need more clarification, subjective opinions, statements that express a value judgment or personal assessment rather than an empirical assertion, are not check-worthy, since they cannot be verified against scientific literature. For example:

✗ **Not check-worthy — Subjective opinion**
Gene editing is one of the most exciting developments in the history of science.

✗ **Borderline — Vague statement (needs more context)**
These approaches lead to increased distractibility.

For each claim, choose one of the following labels:

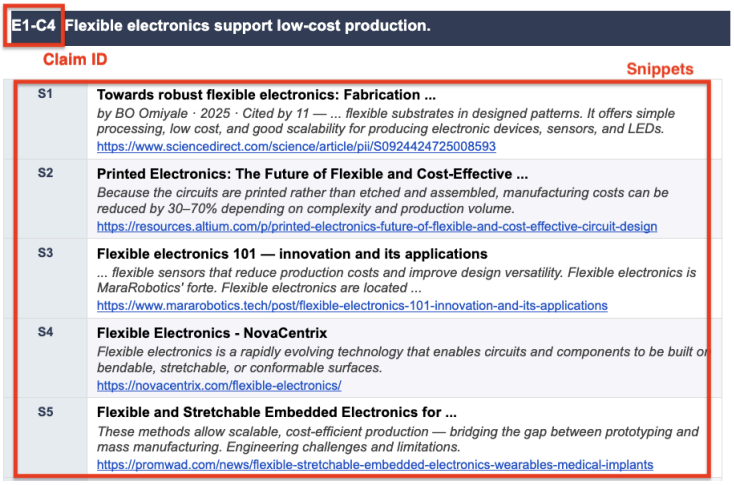
Yes	The claim makes a specific, verifiable scientific assertion that can be (theoretically) fact-checked against scientific literature.
No	The claim cannot be verified against scientific literature.
Not Sure	The claim is in a grey zone: it could be interpreted as check-worthy or not, depending on context.

Important: if you label a claim as No or Not Sure, you can skip the Factuality label for that row.

Label 2: Is This Claim Factual?

Here, you should evaluate whether the claim is supported by the **evidence** we have provided alongside it. There are two evidence types for each scientific concept:

- Search snippets from Google below each claim ID. These are web sources retrieved specifically for that claim.



- A link to the survey paper at the top of the document:



To annotate factuality, first, check the snippets. Note that it's enough to simply read the snippet, there is no need to go to the provided link and search for evidence there.

If you can't find evidence there, check the full paper (no need to read it all, try to Ctrl + F what you are looking for). Don't take too much time searching for evidence. If you can't find any obvious evidence supporting the claim after a few minutes, use the "Cannot Verify" label.

Important:

1. Even though some claims might be obviously correct, make sure they are supported in the given evidence - we want the annotations to be grounded in the evidence and not in your own knowledge.
2. It is not allowed to look up any additional sources for the purpose of this study.
3. If you see a case where you are sure the claim is correct but can't find any supporting evidence, use the "**Comment**" column to let us know, it would be interesting for us to see if there are cases where the provided evidence does not align with your own knowledge!

In some cases, no Google search snippets are available for a claim. If this happens, base your judgment on the survey paper alone.

For each claim, choose one of the following labels:

Yes	At least one source (either one snippet or the survey paper) supports the claim, and none clearly contradict it.
Debatable	The sources conflict. At least one supports and at least one contradicts the claim.
Cannot Verify	None of the sources supports or contradicts the claim.

Not Factual	At least one source contradicts the claim, and none of the sources support it.

Practical Tips

- Work through claims in order, the rows are grouped by scientific concept, so you will stay in context.
- If a claim is ambiguous, annotate it as you would interpret it on a first read. Do not overthink it.
- If you are unsure between two labels, use the comment column to note your reasoning. This will helps us during analysis.
- Expected time: approximately 20 minutes per concept.

Questions?

If anything is unclear or you encounter a case that the guidelines do not cover, please reach out before making a judgment call. We would rather you ask than guess since your annotations feed directly into our analysis, so quality matters more than speed.

Thank you again for your contribution to this research!